

File 71: Creekside Stream Monitoring Field Data Sheet

Field Data Collection Form for Fluvial Geomorphology & Wild Trout Spawning Assessments

Stream Monitoring Field Form Standard: This document represents the official **Creekside Stream Monitoring Field Data Sheet (File 71)** for the **Anderson Creek Showcase Restoration Project (Roya's Cabin Reach)**. It provides Hunter Morris, Trout Unlimited, and university interns (UGA Warnell, Georgia Tech) with a ready-to-print, structured field data sheet to systematically record Wolman pebble counts, EPA RBP benthic macroinvertebrates, physical water parameters, and trout spawning nests (redds).

I. Project & Station Header Information

- **Project Name / Reach:** Anderson Creek Showcase Restoration (Roya's Cabin Reach)
- **Station ID (HUC-8):** GA-03150105 / Station # _____ (e.g. Monument #01)
- **Field Observers / Crew:** _____
- **Sampling Date:** ____ / ____ / 202__ **Time:** ____ : ____ AM/PM
- **Weather Conditions:** ☐ Sunny ☐ Partly Cloudy ☐ Overcast ☐ Raining
- **Air Temperature:** _____ °F **Recent Precipitation (24 hr):** _____ inches

II. Hydrological & Biochemical Water Quality Parameters Log

Equipment Calibrated Prior to Field Work? ☐ Yes ☐ No (YSI ProDSS Calibration Complete)

Parameters	Unit	Reading #1	Reading #2	Reading #3	Arithmetic Mean	Standard Deviation
Water Temperature	°F (°C)					
Dissolved Oxygen	mg/L					
Turbidity Index	NTU					
Specific pH	S.U.					

Parameters	Unit	Reading #1	Reading #2	Reading #3	Arithmetic Mean	Standard Deviation
Stream Flow Velocity	ft/s (m/s)					
Gauge Stage Height	ft (m)					

III. Wolman Pebble Count Tally Grid

Measure 100 random bed substrate particles across the cross-section riffle profile to compute median diameter (D_{50})

Particle Class	Size Range (mm)	Sieve Classification	Tally (e.g. III, IIII, etc.)	Total Count
Silt / Clay	< 0.062 mm	Silt		
Very Fine Sand	0.062 - 0.125 mm	Sand		
Fine Sand	0.125 - 0.25 mm	Sand		
Medium Sand	0.25 - 0.50 mm	Sand		
Coarse Sand	0.50 - 1.0 mm	Sand		
Very Coarse Sand	1.0 - 2.0 mm	Sand		
Very Fine Gravel	2.0 - 4.0 mm	Pebble (Spawning)		
Fine Gravel	4.0 - 5.7 mm	Pebble (Spawning)		
Medium Gravel	5.7 - 8.0 mm	Pebble (Spawning)		
Medium Gravel	8.0 - 11.3 mm	Pebble (Spawning)		
Coarse Gravel	11.3 - 16.0 mm	Pebble (Spawning)		
Coarse Gravel	16.0 - 22.6 mm	Pebble (Spawning)		
Very Coarse Gravel	22.6 - 32.0 mm	Pebble (Spawning)		
Very Coarse Gravel	32.0 - 45.0 mm	Pebble (Spawning)		
Very Coarse Gravel	45.0 - 64.0 mm	Pebble (Spawning)		
Small Cobble	64 - 90 mm	Cobble		
Medium Cobble	90 - 128 mm	Cobble		

Particle Class	Size Range (mm)	Sieve Classification	Tally (e.g. III, IIII, etc.)	Total Count
Large Cobble	128 - 180 mm	Cobble		
Very Large Cobble	180 - 256 mm	Cobble		
Small Boulder	256 - 512 mm	Boulder		
Medium Boulder	512 - 1024 mm	Boulder		
Large Bedrock	> 1024 mm	Bedrock		
TOTAL COUNT	—	—	—	100

IV. EPA RBP Benthic Macroinvertebrate Tally Card

Conduct a standard 3-kick net composite sample in gravel riffles. Sort and count sensitive EPT aquatic insects

1. Group 1: Sensitive Taxa (EPT Index - Indicator of clean, cold trout waters)

- **Mayflies (Ephemeroptera):**
- *Heptageniidae* (Flathead Mayfly): _____ *Baetidae* (Minnow Mayfly): _____ *Ephemerellidae* (Spiny Crawler): _____
- **Stoneflies (Plecoptera):**
- *Perlidae* (Common Stonefly): _____ *Peltoperlidae* (Roach-like Stonefly): _____ *Nemouridae* (Forest Stonefly): _____
- **Caddisflies (Trichoptera):**
- *Hydropsychidae* (Net-spinner): _____ *Glossosomatidae* (Saddlecase): _____ *Limnephilidae* (Northern Case): _____

2. Group 2: Moderately Tolerant Taxa

- **Dragonflies / Damselflies (Odonata):** _____
- **Crayfish (Decapoda):** _____
- **Dobsonflies / Hellgrammites (Corydalidae):** _____
- **Blackflies (Simuliidae):** _____

3. Group 3: Highly Tolerant Taxa (Indicator of siltation and hypoxia)

- **Midges (Chironomidae):** _____
- **Aquatic Worms (Oligochaeta):** _____
- **Leeches (Hirudinea):** _____

4. Biological Health Calculations

- **Total Taxa Richness:** _____ distinct families observed.

- **EPT Index Score:** _____ (Sum of Mayfly + Stonefly + Caddisfly families).
- *Evaluation:* EPT ≥ 12 = Excellent; 8 - 11 = Good; 4 - 7 = Fair; ≤ 3 = Poor.

V. Wild Trout Spawning Nest (Redd) Log Sheet

Walk the riffle meanders in late autumn. Identify and map clean gravel pockets used by spawning trout

Redd #	Geospatial Coordinates (Lat/Lon)	Water Depth (ft)	Flow Velocity over Redd (ft/s)	Redd Length (ft)	Redd Width (ft)	Parent Trout Species (Brown/Brook)
01						
02						
03						
04						
05						
06						
07						
08						

Redd Mapping Field Notes

(Describe any sediment deposits near redds, presence of spawning parent trout, canopy shading levels, or wood debris structures nearby)

Prepared by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin.